

PERMUTATIONS AND COMBINATIONS

SUB-TOPICS

- ⇒ Exponent of Prime in $n!$
- ⇒ Properties.

- ⇒ Permutations Under Restrictions ...
- ⇒ Permutations, Combinations

- ⇒ Methods

- GAP
- Together
- Not all together

$$1! = 1$$

$$2! = 2$$

$$3! = 6$$

$$4! = 24$$

$$5! = 120$$

$$6! = 720$$

$$7! = 5040$$

$$8! = 40320$$

$$9! = 362880$$

$$10! = 3628800$$

- ⇒ Rank of a Word.

- ⇒ Circular Permutations.

- ⇒ Formation of Groups

- ⇒ Deck of Cards.

- ⇒ Divisibility Models.

- ⇒ Circular Permutations

- ⇒ Formation of Groups

- ⇒ Models based on Divisors
 - Sum of Divisors
 - No. of Even/Odd

- ⇒ Geometry Moduls.

- ⇒ Maximum NO. of Points of Intersection

- ⇒ NO. of Polygons Formed by Chess etc ...

Divisors ...

→ Lines

→ Triangles

→ Circles ...

- ⇒ All Possible Selections

- ⇒ Principle of Inclusion and Exclusion

- ⇒ Derangements

Concepts &

"RANK OF WORD"

Key Points

A B C D E F G H I J K
L M N O P Q R S T U V
W X Y Z.

$$1. \frac{n P_r}{n P_{r-1}} = n - r + 1$$

$$2. n P_r = n(n-1)P_{r-1}$$

$$3. n P_r + r \cdot n P_{r-1} = (n+1)P_r$$

$$n P_r = \frac{n!}{(n-r)!}$$

Permutations : Arrangements

Combinations : Collection

: Anagrams $\rightarrow$ Total No. of Permutations - 1

(i) NO. of Ways in n Obj's where P_0 are same, rest all are of Distinct = $\frac{n!}{P_0!} \cdot n P_{P_0}$

* (ii) NO. of Ways of n Obj's where P_1 are one kind
 P_2 are 2nd kind = $\frac{n!}{P_1! P_2! P_3! \dots P_k!}$
 P_k are kth kind

(iii) NO. of Ways of Selection of m Different things Out of n Distinct things is $n C_m$.

(iv) NO. of Ways of Diff Obj taken r at a time where Repetitions is allowed is n^r .

(v) Selecting atleast one thing Out of n things = $2^n - 1$

(vi) n Similar things taken Any NO. of things (Including Nothing) is $n+1$.

(ii) No. of ways of selecting atleast one thing out of n different things is n .

(iii)

P	Similar things of one type
Q	Second
R	Third Type

= NO. of Ways of Selecting atleast one thing
is $(P+1)(Q+1)(R+1)-1$

(ix) Out of n different things if P, Q, R similar things of different type $[(P+1)(Q+1)(R+1)2^n - 1]$

- : In Geometrical Models NO. of Diagonals forms by n Different Points $nC_2 - n$ (Polygon).
- : Three Points is Must to form Both Either Triangle (or) Circle.
- : Words Other than Vowels is Consonants.

Divisibility Modules:

- i. 3 : Sum of its Digits is Divisible by 3.
- ii. 4 : If Last two Digits are Divisible by 4.
- iii. 6 : Both 2 & 3 Divisible.
- iv. 7 : $(2 \times \text{Unit Place Digits}) - (\text{Number Formed by Other})$
- v. 8 : Last three Digits Divisible by 8.
- vi. 9 : Sum of Digits must be 9.
- vii. 11 : $(\text{Sum of Digits in Odd places}) - (\text{Sum of Digits in even places})$

Divisors:

$\begin{cases} \rightarrow \text{Sum of Divisors} \\ \rightarrow \text{Proper / Improper Divisors.} \end{cases}$

No. of Divisors =

$\rightarrow \text{GAP: } (n-m)! (n-m+1) P_m$

$\rightarrow \text{Together: } (n-m+1)! m!$

$\rightarrow \text{Not all things Together: } n! - (n-m+1)! m!$